

Power Budget

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Project Name: CropSCOUT
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Version: 1

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current	Total Current	Unit
	3.3 V switching regulator	LM2575-3.3	3.3- 40V	1	3.2	3.2	A
	5 V switching regulator	LM2575-5	5 - 40V	1	3.2	3.2	A
	ESP32	ESP32-S3-WROOM-N1	3.3V	1	500	500	mA
	SPI motor driver	TLE9201SGAUMA1TR-ND	5V	4	8	32	A
+12V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	ESP32	ESP32-S3-WROOM-N1	3.3V	1	500	500	mA
	SPI motor driver	TLE9201SGAUMA1TR-ND	5V	4	8	32	A
	Subtotal					532	mA
	Safety Margin					25%	
	Total Current Required on +12V Rail					665	mA
8-pin ribbon connector	Shared team power	BL5000C21703S2PFTM106IBC2NN	12.6V	1			
Total Remaining Current Available on +12V Rail						#REF!	mA

+9V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	ESP32	ESP32-S3-WROOM-N1	3.3V	1	500	500	mA
	SPI motor driver	TLE9201SGAUMA1TR-ND	5V	4	8	32	A
						Subtotal	532 mA
						Safety Margin	25%
						Total Current Required on +5V Rail	665 mA
Regulator	3.3 V regulator	LM2575-3.3	5- 40V	1	3.2	3.2	A
	Total Remaining Current Available on +5V Rail					-661.8	mA

+3.3V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	ESP32	ESP32-S3-WROOM-N1	3.3V	1	500	500	mA
	SPI motor driver	TLE9201SGAUMA1TR-ND	5V	4	8	32	A
						Subtotal	532 mA
						Safety Margin	25%
						Total Current Required on +3.3V Rail	665 mA
Regulator	3.3 V regulator	LM2575-3.3	3.3- 40V	1	3.2	3.2	A
	Total Remaining Current Available on 3.3V Rail					-661.8	mA

<i>External Power Source 1</i>	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 1 Selection	Plug-in Wall Supply	N/A	110VAC	+24V	5000	5000	mA
<i>Total Remaining Current Available on External Power Source 1</i>						5000	mA

<i>External Power Source 2</i>	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 2 Selection	BATTERY ALKALINE 9V	2375-6LR61-1B-ND	+9V	-9V	500	500	mA
<i>Total Remaining Current Available on External Power Source 2</i>						500	mA

<i>Calculated Battery Life</i>							
	Component Name	Part Number	Supply Voltage Range		Capacity (mAh)	Required By Regulators	
	Battery	BL5000C21703S2PFTM106IBC2NN	12V		10000	4000	
<i>Battery Life</i>						2.5	hours

Notes:
 External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).
 If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit